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CE F231
Fluid Mechanics

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Lecture 30: Notches and Weirs

Empirical approach: Francis's formula

Francis's formula

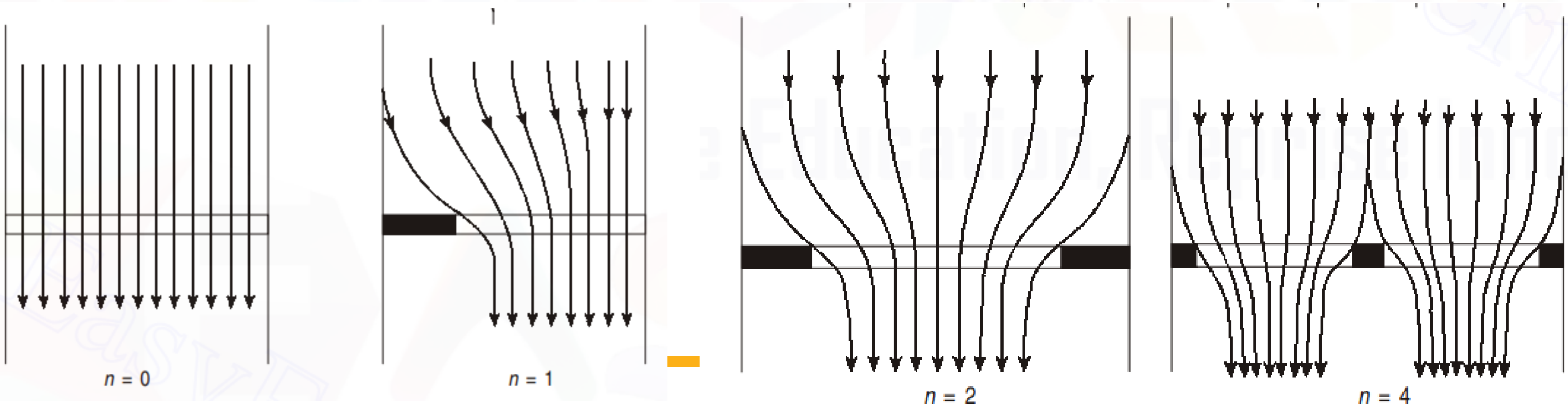
When the velocity of approach is not considered

$$Q = 1.84(L - 0.1\ nH)H^{3/2}$$

With the velocity of approach taken into account

$$Q = 1.84(L - 0.1\,nH_1)\left(H_1^{3/2} - h_a^{3/2}\right)$$

where $H_1 = (H + h_a) = \left(H + \frac{V_a^2}{2g}\right)$, H is the height of water surface above the crest of the weir, and n is the number of end contractions.



Empirical approach: Bazin's formula

Bazin's formula

With the velocity of approach taken into account

$$Q = m_1 \sqrt{2g}\, L H_1^{3/2}$$

in which the coefficient m_1 is given by

$$m_1 = \left(0.405 + \frac{0.003}{H_1}\right)$$

H_1 is in meters (m)

and H_1, which is called the still water head is given by

$$H_1 = \left(H + \alpha \frac{V_a^2}{2g}\right)$$

where α is a constant, the mean value of which has been given by Bazin as 1.6.

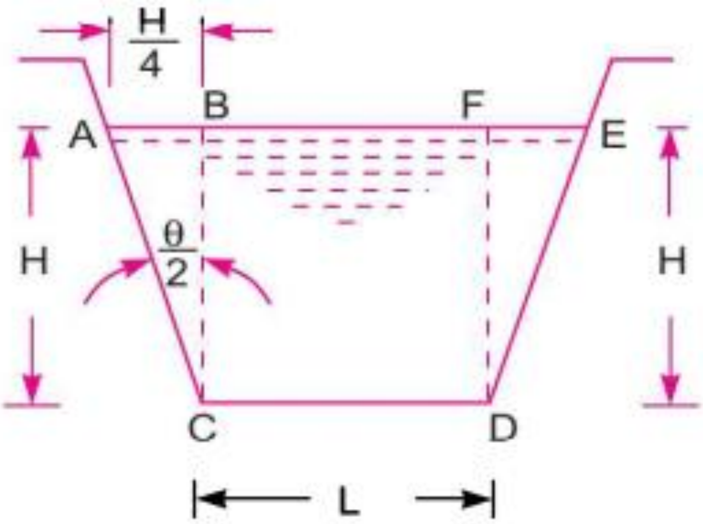
https://pubs.usgs.gov/wsp/0150/report.pdf

Cipolletti weir or notch

Discharge through rectangular weir/notch accounting for end contraction

$$Q = \frac{2}{3} \times C_d \times (L - 0.2\,H)\,\sqrt{2g} \times H^{3/2}$$

$$= \frac{2}{3} \times C_d \times L \times \sqrt{2g}\;H^{3/2} - \frac{2}{15} \times C_d \times \sqrt{2g} \times H^{5/2}$$

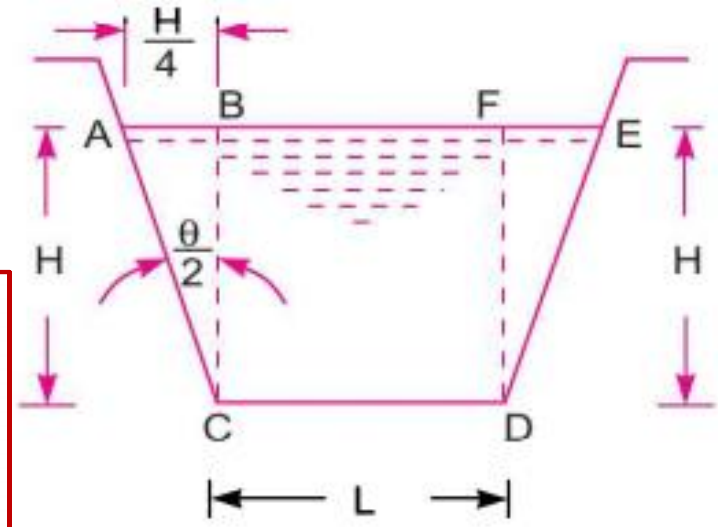


The ends of the weir, instead of being vertical, are inclined outward such that the discharge through the added area counterbalances the decrease from the end contraction

Discharge through triangular weir/notch with $\theta/2 = \tan^{-1}\frac{1}{4} = 14°\,2'$.

$$\frac{8}{15} \times C_d \times \sqrt{2g} \times \tan\frac{\theta}{2}\;H^{5/2} = \frac{2}{15} \times C_d \times \sqrt{2g} \times H^{5/2}$$

Cipolletti weir or notch



Discharge through rectangular weir/notch accounting for end contraction

$$Q = \frac{2}{3} \times C_d \times (L - 0.2\,H)\,\sqrt{2g} \times H^{3/2}$$
$$= \frac{2}{3} \times C_d \times L \times \sqrt{2g}\;H^{3/2} - \frac{2}{15} \times C_d \times \sqrt{2g} \times H^{5/2}$$

Discharge through triangular weir/notch with $\theta/2 = \tan^{-1}\frac{1}{4} = 14°\,2'$.

$$\frac{8}{15} \times C_d \times \sqrt{2g} \times \tan\frac{\theta}{2}\;H^{5/2} = \frac{2}{15} \times C_d \times \sqrt{2g} \times H^{5/2}$$

Thus discharge through cipolletti weir is

$$Q = \frac{2}{3} \times C_d \times L \times \sqrt{2g}\;H^{3/2}$$

If velocity of approach, V_a is to be taken into consideration,

$$Q = \frac{2}{3} \times C_d \times L \times \sqrt{2g}\;[(H + h_a)^{3/2} - h_a^{3/2}]$$

Broad-crested weir

Weir with wide crest → Crest that extends horizontally in the direction of flow far enough to support the nappe so that hydrostatic pressures are fully developed for at least some short distance

0.4 ≤ H/B ≤ 1.6 →Narrow-crested weir

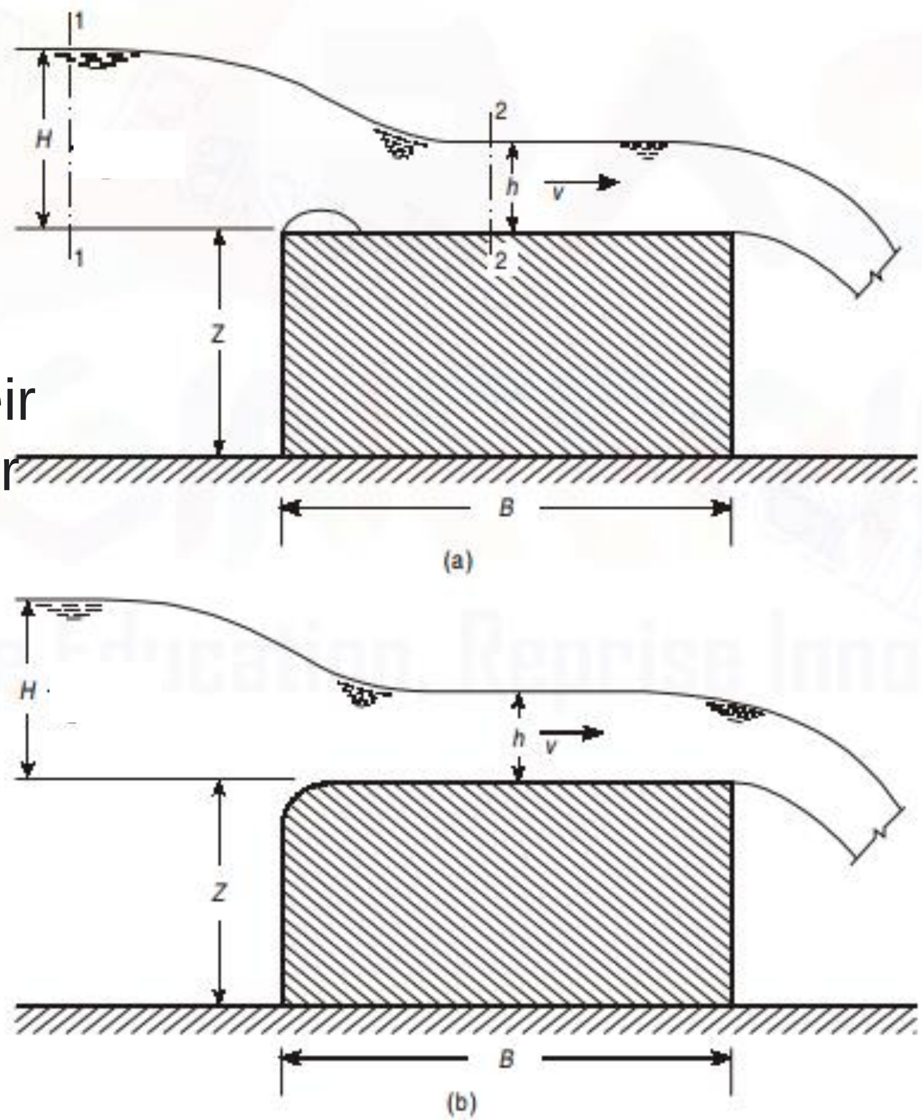
0.1 ≤ H/B ≤ 0.4 → Broad-crested weir

$$H = h + \frac{v^2}{2g}$$

$$v = \sqrt{2g(H-h)}$$

$$Q = C_d L h \sqrt{2g(H-h)}$$

(Without accounting for velocity of approach)



https://docs.bentley.com/

Broad-crested weir

$$Q = C_d L h \sqrt{2g(H-h)}$$

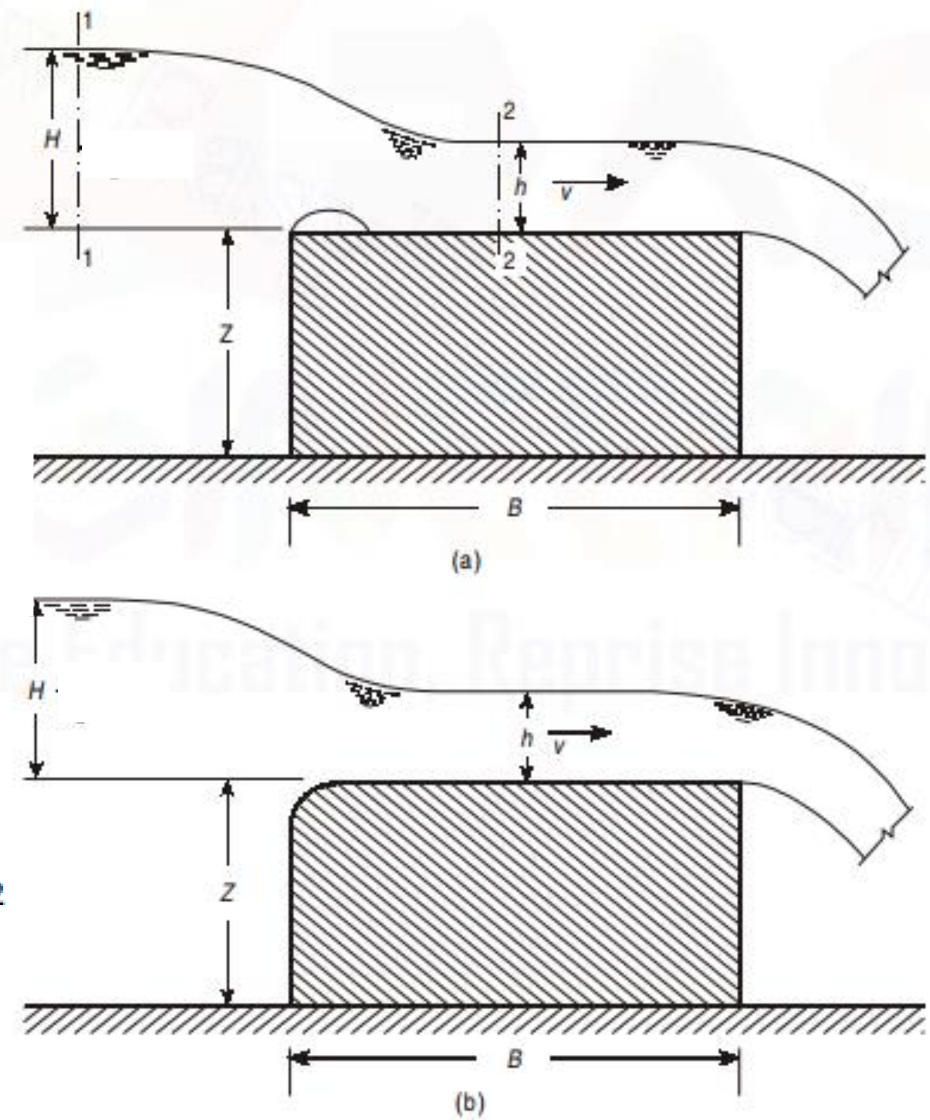
$$\frac{dQ}{dh} = C_d L \sqrt{2g}\left[\sqrt{H-h} - \frac{h}{2\sqrt{H-h}}\right] = 0$$

$$h = \frac{2}{3}H.$$

(Critical depth for maximum discharge)

$$Q = C_d L\left(\frac{2}{3}H\right)\sqrt{2g}\left(\frac{1}{3}H\right)^{1/2} = 1.70\ C_d L H^{3/2}$$

(Without accounting for velocity of approach)

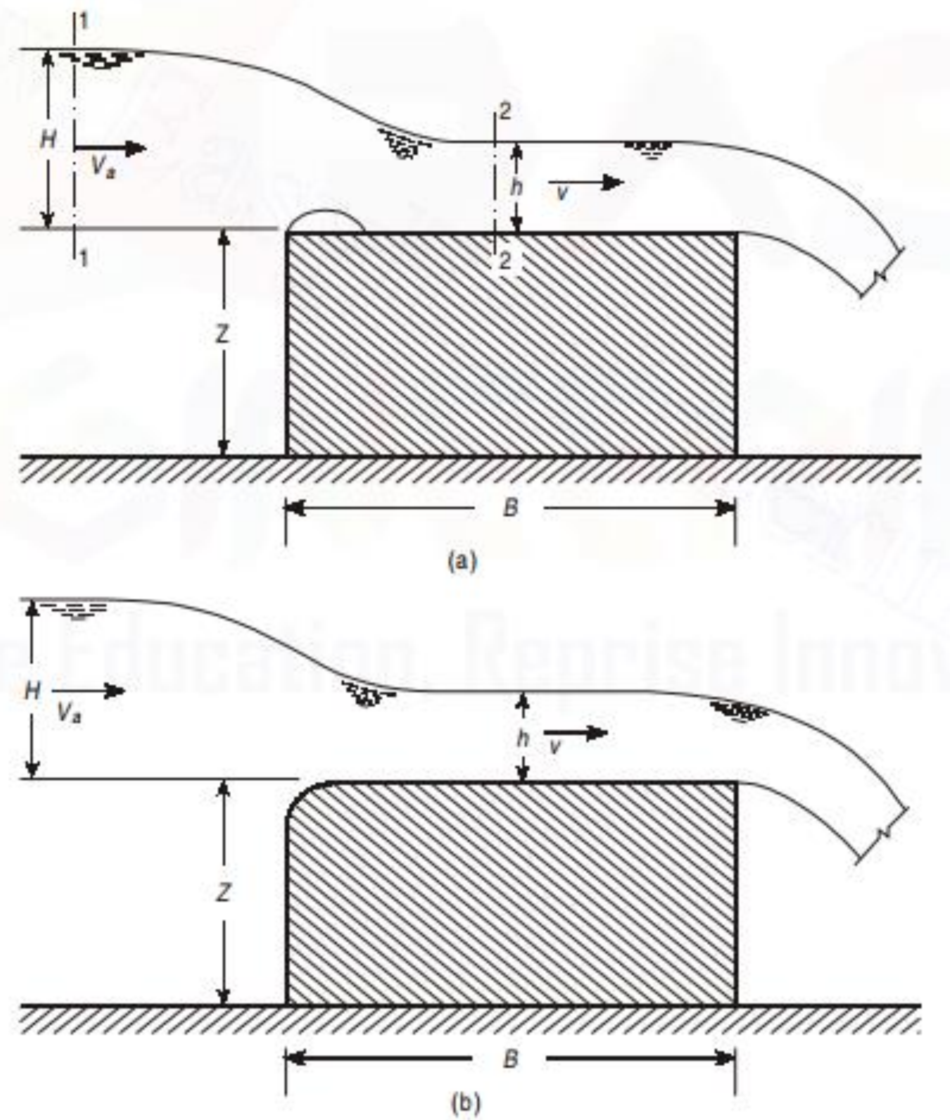


Broad-crested weir

Q_{max} accounting for velocity of approach:

$$Q = 1.70\, C_d L H_1^{3/2}$$

$$H_1 = (H + h_a) = \left[H + \left(\frac{V_a^2}{2g} \right) \right]$$



Submerged weir / drowned weir

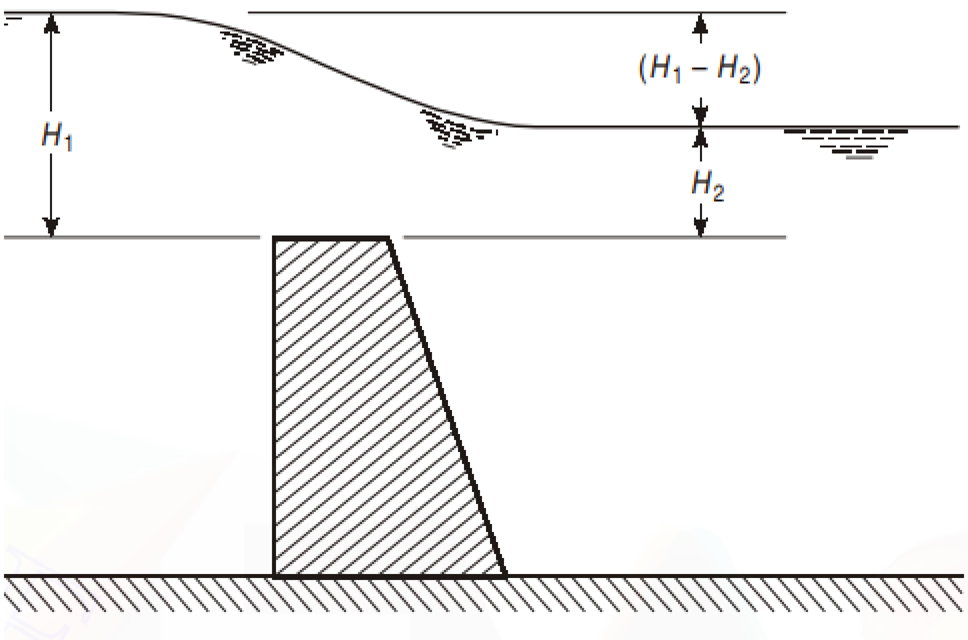
$Q_1 \rightarrow$ Portion b/w U/S and D/S water surface is treated as freely discharging weir

$Q_2 \rightarrow$ Portion between D/S water surface and crest of weir is treated as drowned orifice

$$Q_1 = \frac{2}{3} C_{d_1} L \sqrt{2g} (H_1 - H_2)^{3/2}$$

$$Q_2 = C_{d_2} (L \times H_2) \sqrt{2g(H_1 - H_2)}$$

$$Q = Q_1 + Q_2$$



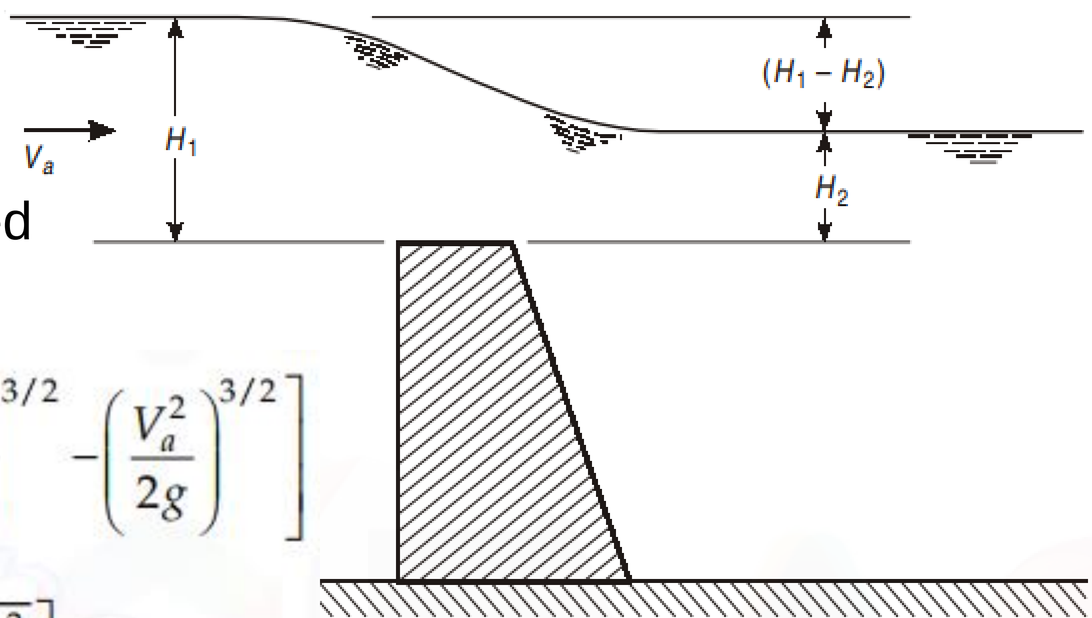
(Without accounting for velocity of approach)

Submerged weir / drowned weir

$Q_1 \rightarrow$ Portion b/w U/S and D/S water surface is treated as freely discharging weir

$Q_2 \rightarrow$ Portion between D/S water surface and crest of weir is treated as drowned orifice

$$Q = Q_1 + Q_2$$



$$Q_1 = \frac{2}{3} C_{d_1} \sqrt{2g}\, L \left[\left\{ (H_1 - H_2) + \frac{V_a^2}{2g} \right\}^{3/2} - \left(\frac{V_a^2}{2g} \right)^{3/2} \right]$$

$$Q_2 = C_{d_2} (L \times H_2) \left[\sqrt{2g(H_1 - H_2) + V_a^2} \right]$$

(Accounting for velocity of approach)

THANK YOU